

1. How many ways can we arrange the letters in the word 'APPEAR'?

There are 6 letters, 2 A's, 2 P's, 1 E, and 1 R.
 We can put the A's in 2 spots $\binom{6}{2}$ ways.
 Then there are 4 spots left, so $\binom{4}{2}$ ways to
 put the P's in. $\binom{2}{1}$ ways to put in the E
 and $\binom{1}{1}$ way to put in R.

$$\binom{6}{2} \cdot \binom{4}{2} \cdot \binom{2}{1} \cdot \binom{1}{1} = 15 \cdot 6 \cdot 2 \cdot 1 = 180 \text{ ways.}$$

2. How many ways can we arrange the letters in the word 'MISSISSIPPI'?

L	#
M	1
I	4
S	4
P	2
total	11

$$\binom{11}{1} \cdot \binom{10}{4} \cdot \binom{6}{4} \cdot \binom{2}{2}$$

$$= 11 \cdot 210 \cdot 15 \cdot 1 = 34,650$$

$$= \frac{11!}{1! \cdot 4! \cdot 4! \cdot 2!} \quad \text{OR} \quad \leftarrow$$

$$\text{OR} = \frac{11!}{1! \cdot 4! \cdot 4! \cdot 2!} \quad \nearrow$$

3. Using the formula for combination, show that

$$\binom{n}{k} = \binom{n}{n-k}$$

For clarity, we will write the formula w/ x's + y's.

$$\binom{x}{y} = \frac{x!}{(x-y)! y!}$$

Now, we will use the formula on the

RHS (right hand side.)

$$\binom{n}{n-k} = \frac{n!}{(n-(n-k))! (n-k)!} = \frac{n!}{(n-n+k)! (n-k)!}$$

$$= \frac{n!}{k! (n-k)!} = \frac{n!}{(n-k)! k!} = \binom{n}{k}$$

4. Give a combinatorial proof (a way to count the same thing in both ways below) that

using Pizza toppings: $\binom{n}{k} = \binom{n}{n-k}$
 If there are n pizza toppings available, we can pick a pizza with k toppings by choosing the k toppings in $\binom{n}{k}$ ways.

OR we could tell them the k toppings by saying the $n-k$ toppings NOT to put on our pizza. $\binom{n}{n-k}$ ways to do this.

Thus, $\binom{n}{k} = \binom{n}{n-k}$. □

5. Give a combinatorial proof (a way to count the same thing in both ways below) that

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n$$

We will keep the Pizza idea, but count all possible toppings. The LHS is the sum of $\binom{n}{k}$ for k from 0 to n . Each of these count the number of ways to order a pizza with k toppings. Since every pizza has some # of toppings, this is all possible toppings.

OR

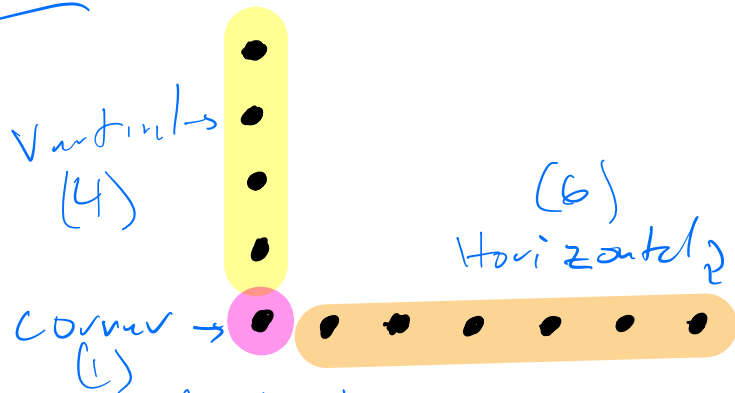
We could count the same total possible pizzas with the RHS by having the cook say each topping and we will respond with a 'yes' or 'no' to each. This is 2 options for each of the n toppings. Which gives 2^n possible pizzas.

Thus $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$. □

From the Book:

Section 1.3: # 6, 12, 14.

#6



How many Triangles?
Note: We exclude



We start by Breaking up our points into 3 groups. The Vertical, the corner, + the horizontal. The corner we do Not include in the other two groups. We will count special cases of Triangles that do Not overlap. Then Add them all!

Case 1: Triangles with the corner, one vertical pt + one horizontal pt.

Note: Every $\triangle$ that has the corner must have one from Both other groups - Not just one group.

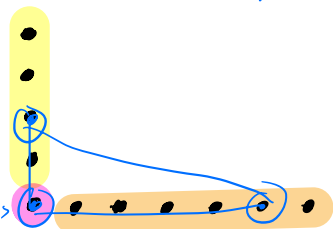
How many $\triangle$'s?

1 corner

$\binom{4}{1}$ options for vertical

$\binom{6}{1}$ options for horizontal

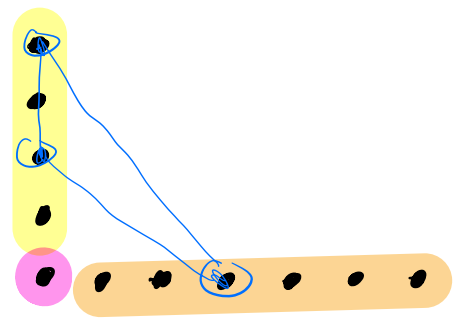
$$1 \cdot \binom{4}{1} \cdot \binom{6}{1} = 4 \cdot 6 = 24 \text{ } \triangle\text{'s!}$$



Case 2: 2 vertical pts and 1 horizontal pt.

$\binom{4}{2}$ options for vertical

$\binom{6}{1}$ options for horizontal

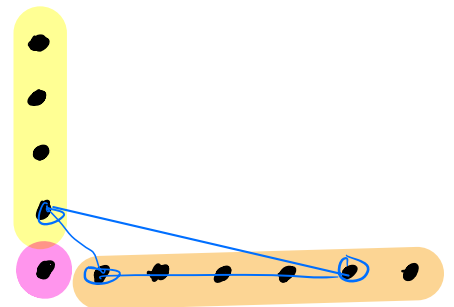


$$\binom{4}{2} \cdot \binom{6}{1} = 6 \cdot 6 = 36 \text{ } \Delta\text{'s}$$

Case 3: 2 horizontal pts and 1 vertical pt

$\binom{6}{2}$ options for the horizontal pts

$\binom{4}{1}$ options for the vertical pt.



$$\binom{6}{2} \cdot \binom{4}{1} = 15 \cdot 4 = 60$$

This covers All the possible triangles So...

Case 1 + Case 2 + Case 3

$$= 24 + 36 + 60$$

$$\boxed{= 120 \text{ triangles}}$$

#12) In book online solution

a) 17^{10} ,

b) $P(17, 10)$

#14) a) Stacking 5 of the 12 distinct chips gives us $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$ possible ways to do this.

$$P(12, 5) = \frac{12!}{(12-5)!} = \frac{12!}{7!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot \cancel{7!}}{\cancel{7!}}$$
$$= 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$$

Yay! the same!

b) $12!$ would be the # of ways to stack all 12 chips. $7!$ here is the number of ways to stack the chips that got stacked after we put our first 5 chips down.

c) We want to divide $12!$ and $7!$ because we want to consider what it means if we pretend we didn't put the last 7 chips down. So those first 5 chips are all that matters and this division will ignore the rest.

d) Yes! with n chips, placing k down in a stack is still

$$n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1) \\ = \frac{n!}{(n-k)!} = P(n, k).$$